



Faculty of Science & Technology
Programme Structure & Syllabus
For
Second Year
Bachelor of Technology
Artificial Intelligence and Machine Learning
Programme Code: BTECAM

Pattern 2024
<With Effect from Academic Year 2025-26>

	Authority	Date
Proposed by	Board of Studies in Artificial Intelligence	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

Title: Programme Structure and Syllabus

Form No: IQAC-101

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Artificial Intelligence
To lead in the development of ethical and inclusive AI systems that promote social good, sustainability, and equitable access to technological advancements

Mission of the Department of Artificial Intelligence	
M1	We aim to empower students and researchers with deep knowledge and practical skills in AI technologies, fostering the next generation of AI experts.
M2	Our goal is developing cutting-edge AI solutions that address global challenges, enhance human capabilities, and drive economic growth through innovative and impactful technologies.
M3	Through collaboration with industry and academia, we strive to push the boundaries of AI, leveraging collective expertise to advance the field and its applications.
M4	We are committed to ensuring the ethical and responsible deployment of AI systems, prioritizing the betterment of society and adhering to the highest standards of integrity and accountability
Values of the Department of Artificial Intelligence	
Pursuing Excellence	Ethics & Social Responsibility
Nurturing Talent	Collaborative Approach
Career Architect	Sustainable Growth Transparency & Accountability
Innovative Outlook	Freedom of Expression

Mapping of Mission Statement of Department to University Mission Statement

Mission Statement	VU-M1	VU-M2	VU-M3	VU-M4
M1	3	3	2	2
M2	1	2	2	3
M3	2	3	2	2
M4	1	1	2	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	To prepare the students to evolve into a professional and committed data science technology workforce, by providing them with a global educational platform with innovative practices.
PEO2	To provide a sufficient mathematical, statistical and computing theory knowledge base, leading to practical and long-lasting data science solutions.
PEO3	To ingrain problem solving skills in data science and sound Engineering principles in students with the help of modern data science tools and techniques.
PEO4	To impart in the students professional and ethical practices based on standard guidelines as applied to the field of artificial intelligence to acquire the right attitude and aptitude for the benefit of society.
PEO5	To guide students for higher education and lead them to avenues of entrepreneurship in areas of artificial intelligence and data science.

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3	PEO4	PEO5
M1	3	3	3	1	2
M2	2	2	3	3	2
M3	3	3	2	2	1
M4	2	1	2	3	2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Outcomes (POs)

PO No.	Statement
PO1	Apply the knowledge of engineering, data science fundamentals and artificial intelligence to solve complex engineering problems.
PO2	Identify, formulate, and analyze complex engineering problems to create solutions using the first principles of mathematics, statistics, and data science.
PO3	Design solutions for system components that meet the specified needs for the benefit of society.
PO4	Apply research methods including design of experiments, statistical analysis and business interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Apply appropriate data mining techniques and statistical tools to model complex engineering activities.
PO6	Apply reasoning using contextual knowledge to assess the needs of society and understand the responsibilities relevant to the engineer specializing in data science.
PO7	Understand the impact of artificial intelligence as applied to the environment, and demonstrate the knowledge for sustainable development.
PO8	Adopt ethical practices as applied to the professional world to execute responsibilities and adhere to norms of the engineering profession.
PO9	Execute professional functions effectively as an individual, as well as a leader or member in diverse multidisciplinary teams.
PO10	Communicate effectively with the engineering community and with society in solving complex artificial intelligence problems in terms of being able to comprehend and write effective reports, make effective presentations, as well as execute and receive clear instructions.
PO11	Demonstrate an ability to lead projects and build cost models in an interdisciplinary professional setting.
PO12	Develop learning mechanisms and inculcate the ability to prepare for lifelong learning in the context of technological change.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PEO1	2	3	2	3	1	2	3	1	1	-	-	2
PEO2	3	3	2	3	1	1	2	1	-	1	1	1
PEO3	2	2	2	2	3	1	3	1	-	1	-	2
PEO4	1	1	1	1	1	1	1	3	3	2	2	-
PEO5	2	2	2	2	1	1	1	1	1	-	2	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Imbibe foundational knowledge of computing theory, statistics, algorithms, and programming paradigms to create innovative, scientific solutions for challenges in the field of data science and artificial intelligence.
PSO2	Prepare for recent trends in the data science industry by adapting to various computing platforms, architectures and changing software paradigms.
PSO3	Evolve new design ideas for real world problems in data science using appropriate mechanisms, design patterns, modeling methods and modern machine learning and statistical tools.
PSO4	Inculcate research attitude and aptitude in students using techniques and standards as applied to data science technology migration and evolution.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3	PSO4
PEO1	2	3	1	2
PEO2	3	2	1	3
PEO3	3	2	3	2
PEO4	2	2	1	1
PEO5	2	2	1	2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty		Science and Technology	Pattern (AY-2025 2026)				2024		
Department		Artificial Intelligence	Date (w.e.f.)				01/07/2025		
Programme		B.Tech (CSE- Artificial Intelligence and Machine Learning) (BTECAM)							
Semester -III									
Multidisciplinary Minor									
Course Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CIE	ESE	Total
PCC	BTECAM24301	Applied Statistical Analysis	3	0	0	3	60	40	100
PCC	BTECAM24302	Applied Statistical Analysis Lab	0	0	2	1	15	10	25
PCC	BTECAM24303	Data Structures	3	0	0	3	60	40	100
PCC	BTECAM24304	Data Structures Lab	0	0	2	1	15	10	25
HSSM	BTECAM24305	Financial Literacy and Banking	2	0	0	2	50	0	50
CEP/FP	BTECAM24306	Project Based Learning - Python	0	0	4	2	30	20	50
VEC	VE24AI01	Environmental Science-I	2	0	0	2	50	0	50
MC	MT24AI01A	Discrete Structures	3	0	0	3	60	40	100
OE	OE24AI01	Ubiquitous Computing	3	0	0	3	60	40	100
Total =24			16	0	8	20	400	200	600

Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit

PCC- Programme Core Course, **HSSM**- Entrepreneurship/Economics/ Management Courses, **CEP/FP**- Comm. Engg. Proje(CEP)/Field Project (FP), **VEC**- Value Education Course, **MC**- Multidisciplinary Minor, **OE**- Open Elective

Honors and Multidisciplinary Minor (Specialization Honors- Blockchain)

Course Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P		CIE	ESE	Total
PCC	BTECAM24301	Applied Statistical Analysis	3	0	0	3	60	40	100
PCC	BTECAM24302	Applied Statistical Analysis Lab	0	0	2	1	15	10	25
PCC	BTECAM24303	Data Structures	3	0	0	3	60	40	100
PCC	BTECAM24304	Data Structures Lab	0	0	2	1	15	10	25
HSSM	BTECAM24305	Financial Literacy and Banking	2	0	0	2	50	0	50
CEP/FP	BTECAM24306	Project Based Learning - Python	0	0	4	2	30	20	50
HONS	BTECAM24H307	Introduction to Cryptocurrency	3	0	0	3	60	40	100
VEC	VE24AI01	Environmental Science-I	2	0	0	2	50	0	50
MC	MT24AI01A	Discrete Structures	3	0	0	3	60	40	100
OE	OE24AI01	Ubiquitous Computing	3	0	0	3	60	40	100
Total=27			19	0	8	23	460	240	700

Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit

PCC- Programme Core Course, **HSSM-** Entrepreneurship/Economics/ Management Courses, **CEP/FP-** Comm. Engg. Proje(CEP)/Field Project (FP), **VEC-** Value Education Course, **MC-** Multidisciplinary Minor, **OE-** Open Elective

Double Minor (Specialization minor-Cybersecurity)

Course Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CIE	ESE	Total
PCC	BTECAM24301	Applied Statistical Analysis	3	0	0	3	60	40	100
PCC	BTECAM24302	Applied Statistical Analysis Lab	0	0	2	1	15	10	25
PCC	BTECAM24303	Data Structures	3	0	0	3	60	40	100
PCC	BTECAM24304	Data Structures Lab	0	0	2	1	15	10	25
HSSM	BTECAM24305	Financial Literacy and Banking	2	0	0	2	50	0	50
CEP/FP	BTECAM24306	Project Based Learning - Python	0	0	4	2	30	20	50
VEC	VE24AI01	Environmental Science-I	2	0	0	2	50	0	50
MC	MT24AI01A	Discrete Structures	3	0	0	3	60	40	100
OE	OE24AI01	Ubiquitous Computing	3	0	0	3	60	40	100
MC	MT24AI01B	Cyber Security and Cyber Laws	3	0	0	3	60	40	100
Total=27			19	0	8	23	460	240	700

Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit

PCC- Programme Core Course, **HSSM**- Entrepreneurship/Economics/ Management Courses, **CEP/FP**- Comm. Engg. Proje(CEP)/Field Project (FP), **VEC**- Value Education Course, **MC**- Multidisciplinary Minor, **OE**- Open Elective

Name of Head of Department

Sign:

Date:

Name of Dean

Sign:

Date:

BTECAM24301:: Applied Statistical Analysis

Course Type	PCC	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

Statistical data analysis is a fundamental course designed to provide students with a comprehensive understanding of statistical techniques and their applications in analysing and interpreting data. The course emphasizes the practical aspects of statistical analysis and equips students with the necessary skills to make informed decisions based on data.

Course Outcomes:

CO No.	Statement
1	Gain an understanding of statistical methods relevant to upper division interdisciplinary courses.
2	Sharpen students' statistical intuition and abstract reasoning as well as their reasoning from numerical data through community-based and other research.
3	Learn and apply the procedure of hypothesis testing.
4	Understand the relationships within data using regression and correlation.
5	Learn advance techniques for statistical analysis.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	3																UN
CO2		3															AP
CO3				3													AP
CO4					3												AP
CO5														3			AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Contents

Unit-1	Introduction to Statistics - Organizing and Graphing Data	Hours 9	CO	BTL
What is statistics? - Types of statistics - Population versus sample - Basic terms - types of variables - cross section verses time series data - different types of data - sources of data -designing studies and collecting good quality dataRaw data- organizing and graphing qualitative data - organizing and graphing quantitative data - Histograms - Cumulative Frequency Distributions - Stem-and-Leaf Display			1	UN
Unit-2	Numerical Descriptive Measures	Hours 9	2	AP
Measures of central tendency for ungrouped data -Measures of dispersion for ungrouped data - Mean, variance, and standard deviation for grouped data - Use of standard deviation - Measures of position - Box and-Whisker Plot				
Unit-3	Discrete Random Variables and Their Probability Distributions	Hours 9	3	AN
Random variables -probability distribution of a discrete random variable -mean of a discrete random variable - standard deviation of a discrete random variable - The binomial probability distribution - The Poisson probability distribution				
Unit-4	Continuous Random Variables & Normal Distribution- Sampling Distributions	Hours 9	4	AP
Continuous probability distribution -The normal distribution - The standard normal distribution - Standardizing a normal distribution - Applications of the normal distribution - determining the z and x values when an area under the normal distribution curve is known -The normal approximation to the binomial distribution. Population and sampling distributions- mean and standard deviation of x-bar - applications of the sampling distribution of x-bar.				
Unit-5	Estimations and Hypothesis Testing	Hours 9	5	AN
Point and interval estimates - estimation of a population mean: (σ known)-estimation of a population mean: (σ not known) Hypothesis tests: Introduction -Hypothesis tests about population mean (σ known) - Hypothesis tests about population mean (σ not known) Chi-Square Tests - The Chi-square distribution - Goodness-of-fit test - Contingency tables - Test of independence or homogeneity - Inferences about the population variance,Analysis of Variance - The F Distribution - One-Way Analysis.				

Text Books

1	<i>Mann Prem S., Lacke Christopher Jay. (2010). Introductory Statistics(Jenn Albanese, Ellen Keohane),7, John Wiley & Sons, Inc</i>
2	<i>Christopher Chatfield. (1994). Statistics for Technology: A Course In Applied Statistics, 3, Taylor & Francis Group.</i>

Reference Books

1	<i>Bruce Peter, BruceAndrew. (2017). Practical Statistics for Data Scientists(Shannon Cutt),1, O'Reilly Media, Inc.</i>
2	<i>EasterlingRobert G. (2015). Fundamentals of Statistical Experimental Design and Analysis(Prachi Sinha Sahay, Jo Taylor, Kathryn Sharples), John Wiley & Sons, Ltd.</i>

BTECAM24302: Applied Statistical Analysis Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	--Hr./Week	Lecture	-	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

1	Introduction to R Language and basic R commands - basic calculations - help - installing packages - datatypes - data structures - vectors - subsetting - logical operators - matrices - calculations with vectors and matrices - lists - data frames
2	Programming in R - control flow - if/else - for loop - functions - Reading in data from different sources
3	Summarizing Data - Use R commands to find following summary measures of the given 3 data sets - mean , median , mode, standard deviation, range, variance, coefficient of variation, minimum, maximum, quartiles , interquartile range (IQR). Compare the data-sets based these descriptive measures.-- Data cleaning -missing values
4	Graphing data - Plot following graphs - frequency distribution, bar-plots, pie chart, histogram, polygon and box-and-whisker plot, scatter-plots, stem-and-leaf plot, (Examples of continuous and discrete data)
5	Probability Distributions - Computing probabilities of an event by applying various probability distribution function viz. binomial distribution , Poisson distribution , normal distribution , Chi-squared distribution , student t distribution , F distribution, Confidence interval computation
6	Hypothesis testing -1 - One sample t-Test - Two sample t-Test , independent samples t-test - non parametric tests - One-way ANOVA - Two-way ANOVA
7	Hypothesis testing -2 - Comparing group means using the independent samples t test , F-test , Chi square test
8	Confidence Interval Estimation - Proportion test , z-test , t-test , confidence

BTECAM24303::Data Structures

Course Type	PCC	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course helps the student to study the representation and applications of Stack and Queue data structures, tree data structure and its traversals. It also focuses on graph data structure and usage of hashing techniques.

Course Outcomes

CO No.	Statement
1	To understand the representation and applications of stack and Queue data structure.
2	To perform various operations on linked lists.
3	To demonstrate the use of binary tree traversals and perform operations on them.
4	To Use advanced trees and its applications for various purposes.
5	To Apply the Graph data structure to solve the applications of it and use hashing techniques.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	3																UN
CO2		3															AP
CO3							2										AP
CO4												3					AP
CO5													3				AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit-1	Stacks and Queues	Hours -9	CO	BTL
Stacks: Fundamentals of stack, Representation and Implementation of stack using arrays, Applications of stack: Decimal to Binary Conversion, reversing a string, Parsing: Well-form parenthesis, Different expressions, conversions and evaluation. Queues: Fundamentals of queue, Representation and Implementation of queue using arrays, Circular queue: representation and implementation, Applications of queue: Josephus Problem, Job Scheduling, Queue Simulation, Categorizing Data, Doubly Ended Queue, Priority queue. Multiple Stacks and Multiple Queues.			1	UN
Unit-2	Linked Lists	Hours -9	2	AP
Dynamic Memory allocation, Concept of linked organization, singly linked list, doubly linked list, circular linked list, Insertion, Deletion, and traversal on above data structures. Representation and manipulations of polynomials using linked lists. Implementation of linked Lists for Stacks and Queues, Generalized Linked List, operations on GLL like copy, Equality.				
Unit-3	Trees	Hours -9	3	AN
Trees: Basic Terminologies, Binary trees, Types of binary trees, Binary tree representation using array and linked list, Creation of a binary tree, Binary Search Trees, creation of binary Search tree, tree traversals (recursive and non-recursive), finding height and counting leaf nodes of a binary search tree (with and without recursion), Deletion of a node from a binary search tree. Finding mirror image of the binary search tree with and without recursion, Printing a tree level wise.				
Unit-4	Advanced Trees	Hours -9	4	AP
Threaded binary trees, Creation and traversal of in-order, pre-order and post-order threaded binary tree, Insertion and deletion of nodes in threaded binary tree, AVL Trees, Creation of Heap Tree and Heap sorting, Huffman tree.				
Unit-5	Graphs	Hours -9	5	AN
Review of basic terminology, Types of graphs: directed, undirected, weighted graph, Representation of graphs using adjacency matrix, adjacency list, Traversals: Depth First and Breadth First, connected components and spanning trees, Kruskal's and Prim's algorithms for minimum spanning tree, Algorithm for shortest path Dijkstra's algorithm.				

Textbooks

1	<i>“Fundamentals of Data Structures in C”, E. Horwitz, S. Sahani, Anderson-Freed, Second Edition, Universities Press.</i>
2	<i>Data structures using C and C++”, Y. Langsam, M.J. Augenstein, A.M.Tanenbaum, Pearson Education, Second Edition</i>
3	<i>“Data structures using C and C++”, YedidyahLangsam, Moshe Jugenstein, Aaron M. Tanenbaum, Second Edition, Pearson Publications</i>

Reference Books

1	<i>“An Introduction to data Structures with applications”, J. Tremblay, P. Sorenson, TMH Publication, 2nd Edition.2.</i>
2	<i>“C and Data Structures”, E. Balaguruswamy, TMH Publication, 2003.</i>
3	<i>“Data Structures and Algorithm Analysis in C++”, M. Weiss, Pearson Education, Second Edition, 2002.</i>
4	<i>“Data Structures using C”, ReemaThareja, Second Edition, Oxford Publications</i>
5	<i>“Data Structures with C”, Schaum’s online series, SaymourLipschutz, McGraw Hill Publications.</i>
6	<i>“Data Structures and Algorithms”, A.V. Aho, J.E. Hopcroft, J.D. Ullman, Pearson Publication</i>
7	<i>“Data Structures and analysis in C”, Mark Allen Weiss, second Edition, Pearson Publication</i>

BTECAM24304:: Data Structure Lab

Teaching Scheme		Credits		Examination Scheme	
Lecture	--Hr./Week	Lecture	-	CIE Marks	15
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	2Hr./Week	Practical/Studio	1	Total Marks	25
Total	2Hr./Week	Total	1		

List of Experiments

	Unit 1: Stacks and Queues
1	Program for different stack applications
2	Program for Expression conversion and evaluation using stack
3	Program for different queue applications
	Unit 2: Linked List
4	Program for different types of linked list implementation
5	Program for Generalized Linked Lists.
	Unit 3: Trees
6	Assignment on Binary and Binary Search tree and its traversals
7	Assignment on BST operations
	Unit 4: Advanced Trees
8	Assignment on Threaded binary tree and heap tree
9	Assignment on AVL tree
	Unit 5: Graphs

10	Assignment on Graph traversals (BFS and DFS)
11	Assignment on min. spanning tree and Dijkstra's algorithm
12	Mini Project

BTECAM24305::Financial Literacy and Banking

Course Type	HSMC	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2Hr./Week	Lecture	2	CIE Marks	50
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	-
Practical/Studio	--Hr./Week	Practical/Studio	-	Total Marks	50
Total	2Hr./Week	Total	2		

Course Description

This course helps the student to learn financial planning & control, practical aspects of Account. Focuses on financial management skills to practical situations.

Course Outcomes

CO No.	Statement
1	Implement financial knowledge in real life related to personal context and business context.
2	Gain a comprehensive understanding of the structure and evolution of the Indian banking system.
3	Understand various intelligent sources for investment by analyzing capital, insurance and risks involved in processing.
4	Explain balance sheet.
5	Acquaint the current financial trends.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1								3									UN
CO2									2								AP
CO3										2							AN
CO4												2					AP
CO5														2			AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit-1	Introduction	Hours 6	CO	BTL
Understanding Basic Accounting Concepts, Golden Rules, Modern approach, Journal Entries, Ledger, Trial Balance – Basic costing concepts.			1	UN
Unit-2	Financial System	Hours 6	2	AP
Overview of Indian Financial System – Financial Market – Listing Regulations – Primary markets and Secondary Markets – Foreign Direct Investments – Foreign Institutional Investments – Mutual Funds.				
Unit-3	Banking System	Hours 6	3	AN
Indian Banking System – its Structure - Functions – Types of Banks – Types of Deposits and Loans – Central Bank and its functions - Methods of credit control – Indian Money Market and Capital Market - Investment Avenues (Types of Investments), Non – Performing Assets - Challenges and Opportunities in Indian Banking System.				
Unit-4	Financial Statements and Budgeting	Hours 6	4	AP
Using Financial Statements and Budgets - Preparation of personal Balance Sheet and its analysis using ratios - Time value of money concept – Capital Budgeting Techniques (Profitability Index, Pay Back Period, NPV etc.) - Insurance their returns and Risk.				
Unit-5	Current Financial Trends	Hours 6	5	AN
Digital payments and UPI – Basics of cryptocurrency and blockchain – Buy Now, Pay Later (BNPL) – Green finance and sustainable investing – Role of fintech and neobanks – Changes in personal finance after COVID – Impact of government policies, inflation, and interest rates.				

Textbooks/ Reference Books / Journal Articles / Weblink

1	<i>Introduction to accounting: S.N Maheshwari</i>
2	<i>Business Finance : Dr. Mahesh Kulkarni and Dr. Suhas Mahajan</i>
3	<i>Indian Financial System : M.Y. Khan</i>
4	https://byjus.com/commerce/class-11-accountancy-chapter-1-introduction-to-accounting/
5	https://www.topperlearning.com/t-s-grewal-solutions/cbse-class-11-commerce-accountancy/financial-accounting

BTECAM24306 :: Project Based Learning - Python

Course Type	CFP	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	--Hr./Week	Lecture	-	CIE Marks	30
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	20
Practical/Studio	4Hr./Week	Practical/Studio	2	Total Marks	50
Total	4Hr./Week	Total	2		

Course Description

This course provides an introduction to programming and the Python language with core programming concepts like data structures, conditionals, loops, variables, and functions. This course includes an overview of the various tools available for writing and running Python and coding quickly. It also provides hands-on coding exercises using commonly used data structures, writing custom functions, and reading and writing to files. It also introduces to advanced Python programming concepts.

Course Outcomes

CO No.	Statement
1	To understand basic concepts in Python
2	To develop applications employing different object-oriented concepts
3	To implement linear algebra functions for arrays using NumPy
4	To demonstrate high-performance data manipulation and analysis using Pandas Library
5	To plot data visualization using Matplotlib and Seaborn libraries.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1			3														UN
CO2					3												AP
CO3						2											AN
CO4							2										AP
CO5														2			AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's TaxonomyLevel

List of Experiments

1	Create variables of different data types (integer, float, string, boolean). Perform basic operations on them and print the results along with their types using the type() function.
2	Create an array, perform various operations such as insertion, deletion, searching for elements, and iterating through the array. Compare the performance of arrays with lists for certain operations.
3	Write a program to read data from a text file, process the data (such as counting words, lines, and characters), and write the processed data to a new file. Handle potential exceptions that may occur during file operations.
4	Create a program that raises and handles various types of exceptions (such as ValueError, TypeError, and FileNotFoundError). Define and raise custom exceptions to handle specific error conditions in your program.
5	Write a program in Python to demonstrate the concept of a Constructor
6	Write a program in Python to demonstrate the concept of Interface
7	Write a program in Python to demonstrate the concept of Inheritance.
8	Write a program to install NumPy and create arrays using lists and built-in functions. Display the created arrays and their attributes
9	Write a program to create a NumPy array and reshape it into different dimensions. Transpose of the reshaped array and demonstrate swapping axes.
10	Write a program to create two NumPy arrays and perform element-wise addition, subtraction, multiplication, and division. Use universal functions to compute square root, logarithm, and exponential of array elements.
11	Write a program to demonstrate various indexing techniques (e.g., accessing specific elements, rows, columns) and slicing operations to extract subarrays. Include examples of Boolean and fancy indexing.
12	12 Write a program to create a NumPy array and calculate its mean, median, standard deviation, and variance. Use advanced statistical methods to compute percentiles and correlation coefficients.
13	Write a program to read data from a CSV file using Pandas. Display the first few rows (head()), information (info()), and basic statistics (describe()) of the DataFrame. Save the modified DataFrame to a new CSV file.

14	Write a program to create a DataFrame from a dictionary of lists. Use methods like head(), tail(), info(), and describe() to explore and summarize the DataFrame.
15	Write a program to group data in a DataFrame by a specific column using groupby() and compute aggregate statistics (e.g., sum, mean) on grouped data.
16	Write a program to filter rows from a DataFrame based on a condition using boolean indexing or the query() method. Sort the filtered DataFrame by a specific column using sort_values().
17	Write a program to handle missing values in a DataFrame using fillna() or dropna() methods. Perform aggregation operations (sum(), mean(), etc.) on columns of the DataFrame.
18	Write a program to create a simple line plot, scatter plot, histogram, and bar plot using Matplotlib. Customize each plot with appropriate labels, titles, colors, and markers.
19	Write a program to create a box and whiskers plot and an area plot using Matplotlib. Include multiple subplots in a single figure to display different plots side by side.
20	Write a program to create a scatter plot, histogram, bar plot, box plot, and violin plot using Seaborn. Visualize correlations between multiple variables using the pair plot function.

VE24AI01: Environmental Science I

Course Type	VEC	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2Hrs./Week	Lecture	2	CIE Marks	50
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	00
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	50
Total	2 Hrs./Week	Total	2		

Course Objectives

This course explores the fundamental principles of environmental science, including ecosystems, biodiversity, pollution, and sustainable development. Students will examine the impact of human activities on the environment and evaluate solutions for current environmental challenges.

Course Outcomes

CO No.	At the end of this course students will be able to
1	Understand scope of Environmental science, Study different kinds of Ecosystems
2	Gather information about different environmental resources
3	Understand concept of sustainability and sustainable development
4	Understand requirement and meaning of Biodiversity and threats and challenges to maintain it
5	Understand severity of pollution and develop thought process to reduce it

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3		2			3							3	3			UN
CO2				3	3				2							3	AP
CO3							3	3				2			3		AN
CO4		3							3							3	AP
CO5		3								3	3		3		3		AN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit -1	Scope of environmental studies and Ecosystem	Hours 6	CO	BTL
Multidisciplinary nature of environmental studies; Scope and importance; What is an ecosystem? Structure and function of ecosystem; Energy flow in an ecosystem: food chain, food web and ecological succession. Case studies of the following ecosystems: a) Forest ecosystem b) Grassland ecosystem c) Desert ecosystem d) Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)			1	UN
Unit-2	Environmental Recourses	Hours 6	2	AP
Land Resources and land use change; Land degradation, soil erosion and desertification. • Deforestation: Causes and impacts due to mining, dam building on environment, forests, biodiversity and tribal populations. • Water: Use and over-exploitation of surface and ground water, floods, droughts, conflicts over water (international & inter-state). • Heating of earth and circulation of air; air mass formation and precipitation. • Energy resources: Renewable and non-renewable energy sources, use of alternate energy sources, growing energy needs, case studies.				
Unit-3	Sustainability	Hours 6	3	AN
Concept of sustainability and sustainable development with case studies relevant to computer industry				
Unit-4	Biodiversity	Hours 6	4	AP
Levels of biological diversity :genetic, species and ecosystem diversity; Biogeography zones of India; Biodiversity patterns and global biodiversity hot spots • India as a mega-biodiversity nation; Endangered and endemic species of India • Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts, biological invasions; Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity. • Ecosystem and biodiversity services: Ecological, economic, social, ethical, aesthetic and Informational value.				
Unit-5	Pollution	Hours 6	5	AN
Environmental pollution : types, causes, effects and controls; Air, water, soil, chemical and noise pollution • Nuclear hazards and human health risks • Solid waste management: Control measures of urban and industrial waste. • Pollution case studies. Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture.				

Textbooks

1	<i>Morgen, M.D. Morgen J.M. and Wiersima J.H. 1993, Environmental Science: Managing Physical and Biological Resources Wm C Brown Publishers London.</i>
2	<i>Dhaliwal G.S., Sangha G.S. and Ralhan P.K. 2000, Fundamentals of Environmental Science, Kalyani Publishers, New Delhi.</i>
3	<i>Khopkar, S.M. 1993 : Environmental Pollution Analysis, Wiley Eastern Limited New York.</i>
4	<i>Dr. Y. K. Singh, 2006, Environmental Science, New age International Publishers</i>

Reference Books / Journal Articles / Weblink

1	<i>De, A.K. 1990, Environmental Chemistry, Wiley Eastern Ltd., New Delhi.</i>
2	<i>Sodhi G.S. 2005, Fundamentals of Environmental Chemistry : Narosa Publishing House, New Delhi.</i>
3	<i>Morgen, M.D. Morgen J.M. and Wiersima J.H. 1993, Environmental Science Managing Physical and Biological Resources Wm C Brown Publishers London.</i>
4	<i>Caralyn Zehnder, Kalina Manoylov, Samuel Mutiti, Christine Mutiti, Allison VandeVoort, 2018, Introduction to Environmental Science (edition 2), Springer</i>

MT24AI01A::Discrete Structures

Course Type	MC	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

The course will enable student understand, explain and apply the foundational Mathematical Concepts at the core of Computer Science. It will also enable Students to Model Problem Using Graph & Trees, to Understand set, function and relational models to understand Practical Examples and Interpret the associated Operations and Terminologies in Context, to acquire the knowledge of logic and proof techniques, To Learn how abstract algebra is used in coding theory.

Course Outcomes

CO No.	Statement
1	Formulate problems precisely, solve the problems, apply formal proof techniques, and explain the reasoning clearly.
2	Apply appropriate mathematical concepts and skills to solve problems in both familiar and unfamiliar situations including those in real-life contexts.
3	Design and analyze real world engineering problems by applying set theory, propositional logic and to construct proofs using mathematical induction.
4	Specify, manipulate and apply equivalence relations; construct and use functions and apply these concepts to solve new problems.
5	Model and solve computing problem using tree and graph and solve problems using appropriate algorithms.
6	Analyze the properties of binary operations, apply abstract algebra in coding theory and evaluate the algebraic structures.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	1	1	2	1													UN
CO2	1	2	-	2													AP
CO3	2	1	2	1													AN
CO4	1	2	-	2	3												AP
CO5	-	2	1	2									3				AN
CO6	1	2	2	-													AP

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit-1	Set Theory and Logic	Hours 9	CO	BTL
Introduction and significance of Discrete Mathematics Sets – Naïve Set Theory (Cantorian Set Theory), Axiomatic Set Theory, Set Operations, Cardinality of set, Principle of inclusion and exclusion. Types of Sets – Bounded and Unbounded Sets, Diagonalization Argument, Countable and Uncountable Sets, Finite and Infinite Sets, Countably Infinite and Uncountably Infinite Sets, Power set Propositional Logic - logic, Propositional Equivalences, Application of Propositional Logic- Translating English Sentences, Proof. EXEMPLER/CASE STUDIES : Know about the great philosophers- Georg Cantor, Richard Dedekind and Aristotle			1	UN
Unit-2	Relations & Functions	Hours 9	2	AP
Relations and their Properties, n-ary relations and their applications, Representing relations, Closures of relations, Equivalence relations, Partitions, Transitive closure and Warshall 's algorithm. Functions - Surjective, Injective and Bijective functions, Identity function, Partial function, Invertible function, Constant function, Inverse functions and Compositions of functions, The Pigeonhole Principle. EXEMPLER/CASE STUDIES: Know about the great philosophers-Dirichlet				
Unit-3	Graph Theory	Hours 9	3	AN

Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, the handshaking lemma. Single source shortest path-Dijkstra's Algorithm, Planar Graphs, Graph Colouring. EXEMPLER/CASE STUDIES: Google Map		
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Unit-4	Trees	Hours 9	4	AP
Introduction, properties of trees, Binary search tree, tree traversal, decision tree, prefix codes and Huffman coding, cut sets Spanning Trees and Minimum Spanning Tree = Kruskal 's and Prim 's algorithms, The Max flow- Min Cut Theorem (Transport network). EXEMPLER/CASE STUDIES: Tic-Tac-Toe Game Tree				
Unit-5	Algebraic Structures and Coding Theory	Hours 9	5	AN
The structure of algebra, Algebraic Systems, Semi Groups, Monoids, Groups, Homomorphism and Normal Subgroups, and Congruence relations, Rings, Integral Domains and Fields, Coding theory EXEMPLER/CASE STUDIES: Cryptography used in World War II				

Textbooks

1	Kenneth H. Rosen, "Discrete Mathematics and its Applications", Tata McGraw-Hill, ISBN 978- 0-07-288008-3
2	Bernard Kolman, Robert C. Busby and Sharon Ross, "Discrete Mathematical Structures", Prentice-Hall of India /Pearson, ISBN: 0132078457, 9780132078450.

Reference Books / Journal Articles / Weblink

1	Narsingh Deo, "Graph with application to Engineering and Computer Science", Prentice Hall of India, 1990, 0 – 87692 – 145 – 4.
2	Eric Gossett, "Discrete Mathematical Structures with Proofs", Wiley India Ltd, ISBN:978-81- 265-2758-8.
3	Sriram P.and Steven S., "Computational Discrete Mathematics", Cambridge University Press, ISBN 13: 978-0-521-73311-3.

OE24AI01:: Ubiquitous Computing

Course Type	OE	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

Ubiquitous computing represents a paradigm shift in computing where technology seamlessly integrates into everyday environments, enhancing user interaction and providing pervasive computing capabilities. This course explores the foundational theories, principles, and practical applications of ubiquitous computing, emphasizing its relevance in contemporary technological landscapes.

Course Outcomes

CO No.	Statement
1	Understand the fundamental concepts and evolution of ubiquitous computing.
2	Analyze key technologies and architectures used in ubiquitous computing systems.
3	Describe the significance of actuators and controllers in real time application design
4	Use the concept of HCI to understand the design of automation applications.
5	Classify Ubicomp privacy and explain the challenges associated with Ubicomp privacy.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO 2	PSO 3	PSO 4	
CO 1		2											3				UN
CO 2		3												3			AN
CO 3			3												3		AP
CO 4										2						3	AP
CO 5								2					2				EV

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit-1	Introduction to Ubiquitous Computing	Hours 9	CO	BT L
Concept of Ubiquitous Computing and Advantages, Ubiquitous Computing Applications and Scope, Properties of Ubiquitous Computing, Modelling the Key Ubiquitous Computing Properties. Ubiquitous System Environment Interaction. Architectural Design for UbiCom Systems: Smart DEI Model.			1	UN
Unit-2	Ubiquitous Computing Smart Devices and Services	Hours 10	2	UN
Smart Devices and Service properties, Smart mobile, devices and Users, Mobile code, Smart Card Devices and Networks, Service Architecture Models. Service Provision Life-Cycle. Virtual Machines and Operating Systems, OS for Mobile Computers and Communicator Devices.				
Unit-3	Actuation and Control	Hours 10	3	AP
Tagging the Physical World, Sensors and Networks, Micro-Electro-Mechanical Systems (MEMS), Embedded Systems and Real-Time Systems. Programmable and PID type control system, Robots.				
Unit-4	Human Computer Interaction	Hours 7	4	AN
User Interfaces and Interaction for devices, Abstract user interface through Basic Smart Wearable and Implanted Devices. Human- Centered Design (HCD). User Models: Direct and indirect user input and modelling, modelling users' planned tasks and multiple tasks-based computing.				
Unit-5	Ubiquitous Computing Privacy	Hours 9	5	EV
Ubiquitous computing privacy definition, Solove's taxonomy of privacy, legal background, Interpersonal privacy, Ubicomp challenges to privacy: Collection scale, manner and motivation, data types, data accessibility; Case study of privacy solution such as Protecting RFID tags, ways of addressing privacy in UbiComp.				

Textbooks

1	<i>Stefan Poslad, Ubiquitous Computing, Wiley, Student Edition, ISBN:9788126527335 John Krumm, Ubiquitous Computing Fundamentals</i>
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Reference Books / Journal Articles / Weblink

1	<i>Yin-Leng Theng and Henry B. L. Duh, Ubiquitous Computing, IGI, 2nd Edition, ISBN: 9781599046938</i>
2	<i>Adam Greenfield, Everywhere the Drawing age of Ubiquitous Computing, AIGA, 1st Edition , ISBN: 9780321384010</i>
3	<i>Laurence T. Yeng, Evi Syukur and Seng W. Loke, Handbook on Mobile and Ubiquitous Computing, CRC, 2nd Edition, ISBN: 9781439848111</i>

BTECAD24H307:: Introduction to Cryptocurrency

Course Type	Hons	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course introduces the fundamental concepts, technologies, and applications of cryptocurrencies. Students will gain a comprehensive understanding of the history, mechanics, and implications of cryptocurrencies, with a focus on Bitcoin and Ethereum. The course is designed to provide foundational knowledge necessary for the subsequent course on Blockchain Technology, emphasizing current industry needs and trends.

Course Outcomes

CO No.	Statement
1	Identify and describe the fundamental concepts and historical development of cryptocurrencies and Blockchain technology.
2	Explain the mechanics, functionality, and applications of Bitcoin and Ethereum.
3	Apply cryptographic principles to illustrate the workings and security of cryptocurrencies.
4	Evaluate the regulatory frameworks, legal considerations, and ethical issues surrounding cryptocurrencies.
5	Develop a foundational understanding that facilitates advanced study and application of Blockchain technology in various industries.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		2											3				RM
CO 2							2							3			UN
CO 3	3		2														AP
CO 4		2				2		3									AN
CO 5	3		3	2												2	AP

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit-1	Introduction to Cryptocurrencies	Hours 9	CO	BTL
Introduction: Definition and Types of Cryptocurrencies, Overview of Blockchain, Technology History and Evolution of Cryptocurrencies : Digital Currencies before Bitcoin, The Birth of Bitcoin: The Genesis Block and Satoshi Nakamoto, Evolution of Cryptocurrencies, Post Bitcoin Key Cryptocurrency Concepts: Digital Wallets and Keys, Transactions and Ledger Management			1	RM
Unit-2	Fundamentals of Blockchain Technology	Hours 10	2	UN
Blockchain Architecture and Components: Terminology, Structure of a Block, Components of Block, Linking Blocks in the Blockchain, Merkle Trees, Simplified Payment Verification (SPV), Decentralization and Distributed Ledgers, Consensus Mechanisms: Proof of Work (PoW), Proof of Stake (PoS) Blockchain Use Cases and Challenges: Advantages and Disadvantages, Real-world Applications of Blockchain Technology				
Unit-3	Bitcoin and Ethereum	Hours 10	3	AP
Bitcoin Mechanics : What is Bitcoin, History of Bitcoin, Bitcoin Protocol and Network, Bitcoin Mining and Transactions, Security Features of Bitcoin Ethereum Fundamentals : Introduction to Ethereum, Ethereum Virtual Machine (EVM), Smart Contracts and Decentralized Applications (DApps) Comparative Study : Differences and Similarities between Bitcoin and Ethereum, Case Studies				
Unit- 4	Cryptographic Foundations	Hours 7	4	AN
Basics of Cryptography: Introduction to Cryptography, Symmetric and Asymmetric Encryption Public Key Cryptography: Key Generation and Management, Digital Signatures Hash Functions: Properties and Uses, Cryptographic Hash Functions in Blockchain Security and Privacy: Security Features in Cryptocurrencies, Privacy Concerns and Solutions, Double-spending problem.				
Unit-5	Applications, Regulation, and Future Directions	Hours 9	5	AP
Applications of Cryptocurrencies : Cryptocurrencies as Payment Systems, Initial Coin Offerings (ICOs) and Tokenization, Emerging Trends: DeFi, NFTs, and Beyond Regulatory and Ethical Issues : Current Regulatory Landscape, Legal and Tax, Implications, Ethical Considerations and Challenges Future Directions : Innovations in Blockchain and Cryptocurrencies, Potential Future Developments and Opportunities, Industry Needs and Career Opportunities in Cryptocurrency and Blockchain				

Textbooks

1	<i>“Mastering Bitcoin, PROGRAMMING THE OPEN BLOCKCHAIN”, 2nd Edition by Andreas M. Antonopoulos, June 2017, O'Reilly Media, Inc. ISBN: 9781491954386.</i>
2	<i>“Blockchain Applications: A Hands-On Approach”, by ArshdeepBahga, Vijay Madisetti, Paperback – 31 January 2017.</i>
3	<i>“Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction”, July 19, 2016, by Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder, Princeton University Press.</i>

Reference Books / Journal Articles / Weblink

1	<i>“Mastering Blockchain”, by Imran Bashir, Third Edition, Packt Publishing</i>
2	<i>“Mastering Ethereum: Building Smart Contracts and Dapps Paperback” by Andreas Antonopoulos, Gavin Wood, Publisher(s): O'Reilly Media</i>
3	<i>“Blockchain revolution: how the technology behind bitcoin is changing money, business and the world \$ don tapscott and alex tapscot, portfolio penguin, 856157449</i>

MT24AI01B:: Cyber Security and Cyber Laws

Course Type	MC	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course covers fundamental concepts of cybersecurity, including issues and challenges, privacy, and security concerns on social media, and procedures for reporting inappropriate content and understanding legal aspects. It delves into the basic concepts of e-commerce and digital payments, recognizing various transaction types (B2B, B2C, C2C, C2B), familiarizing with different digital payment modes (credit/debit cards, mobile wallets, UPI, etc.), and emphasizing the importance of cybersecurity and RBI guidelines. Additionally, the course teaches preventive measures against digital payment frauds and basic security aspects related to computers and mobile devices, using tools and technologies to protect them, and understanding the nature of cyber-crimes, legal remedies, and reporting procedures.

Course Outcomes

CO No.	Statement
1	Understand the concept of Cyber security and issues and challenges associated with it.
2	To appreciate various privacy and security concerns on online Social media and understand the reporting procedure of inappropriate content, underlying legal aspects and best practices for the use of Social media platforms.
3	Understanding the basic concepts of e-commerce and digital payments involves recognizing various types of e-commerce transactions (B2B, B2C, C2C, C2B), familiarizing oneself with different digital payment modes (credit/debit cards, mobile wallets, UPI, etc.), grasping the importance of cybersecurity and RBI guidelines, and learning preventive measures against digital payment frauds.
4	Understand the basic security aspects related to Computer and Mobiles. They will be able to use basic tools and technologies to protect their devices.
5	To understand the cyber-crimes, their nature, legal remedies and as to how report the crimes through available platforms and procedures.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO 1	3																UN
CO 2		3															UN
CO 3			3														AP
CO 4					3												AP
CO 5													3				UN

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's TaxonomyLevel

Course Content

Unit-1	Introduction to Cyber security	Hours 9	CO	BTL
Defining Cyberspace and Overview of Computer and Web-technology, Architecture of cyberspace, Communication and web technology, Internet, World wide web, Advent of internet, Internet infrastructure for data transfer and governance, Internet society, Regulation of cyberspace, Concept of cyber security, Issues and challenges of cyber security. Cyber Security Key Objectives: CIA triad Cyber Security Principles: Defense in depth, Least privilege, Security by design			1	UN
Unit-2	Social Media Overview and Security	Hours 10	2	UN
Introduction to Social networks. Types of Social media, Social media platforms, Social media monitoring, Hashtag, Viral content, Social media marketing, Social media privacy, Challenges, opportunities and pitfalls in online social network, Security issues related to social media, Flagging and reporting of inappropriate content, Laws regarding posting of inappropriate content, Best practices for the use of Social media, Case studies.				
Unit-3	E-Commerce and Digital Payments	Hours 10	3	AP
Definition of E- Commerce, Main components of E-Commerce, Elements of E-Commerce security, E-Commerce threats, E-Commerce security best practices, Introduction to digital payments, Components of digital payment and stake holders, Modes of digital payments Banking Cards, Unified Payment Interface (UPI), e-Wallets, Unstructured Supplementary Service Data (USSD), Aadhar enabled payments, Digital payments related common frauds and preventive measures. RBI guidelines on digital payments and customer protection in unauthorized banking transactions. Relevant provisions of Payment Settlement Act,2007,				
Unit-4	Digital Devices Security , Tools and Technologies for Cyber Security	Hours 7	4	AP

End Point device and Mobile phone security, Password policy, Security patch management, Data backup, Downloading and management of third party software, Device security policy, Cyber Security best practices, Significance of host firewall and Ant virus, Management of host firewall and Anti-virus, Wi-Fi security, Configuration of basic security policy and permissions.				
Unit-5	Cyber crime and Cyber law	Hours 9	5	UN
Classification of cyber crimes, Common cyber crimes- cyber crime targeting computers and mobiles, cyber crime against women and children, financial frauds, social engineering attacks, , Cybercriminals modus-operandi , Reporting of cyber crimes, Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber crime and offences, Organisations dealing with Cyber crime and Cyber security in India, Case studies. Intellectual Property Rights (IPR) in Cyberspace: Copyright, trademark, and patent laws Digital rights management. Data Protection and Privacy Laws: Digital Personal Data Protection Act (DPDPA), General Data Protection Regulation (GDPR), Health Insurance Portability and Accountability Act (HIPAA)				

Textbooks

1	<i>Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd. (First Edition, 2011)</i>
2	<i>Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform. (Pearson , 13th November, 2001)</i>

Reference Books / Journal Articles / Weblink

1	Cyber Crime Impact in the New Millennium, by R. C Mishra , Auther Press Edition 2010.
2	<i>Electronic Commerce by Elias M. Awad, Prentice Hall of India Pvt Ltd.</i>
3	<i>Cyber Laws: Intellectual Property & E-Commerce Security by Kumar K, Dominant Publishers.</i>
4	<i>Network Security Bible, Eric Cole, Ronald Krutz, James W. Conley, 2nd Edition, Wiley India Pvt. Ltd.</i>
5	<i>Fundamentals of Network Security by E. Maiwald, McGraw Hill.</i>